

Research Areas

- Strongly correlated quantum matter
- Excitonic insulators and condensates
- Two-dimensional materials
- Low-temperature optics and transport

Academic Appointments

2026 – **Stanford Science Fellow**, Stanford University

Incoming fellow starting July 2026

Faculty host: Prof. Tony F. Heinz

Education

2020 – 2026 **Ph.D. in physics**, University of California, Berkeley

Advisor: Prof. Feng Wang

Thesis: Correlated exciton physics in van der Waals electron-hole bilayers

2020 – 2022 **M.A. in physics**, University of California, Berkeley

2016 – 2020 **B.S. in physics**, Peking University (PKU)

GPA: 3.9/4; Rank: 1/200⁺

Publications

40⁺ research articles; 1000⁺ citations. Full publication record:  [Google Scholar](#).

†: equal contribution; ‡: corresponding author.

Selected publications

1. **Ruishi Qi**^{†‡}, Qize Li[†], Jiahui Nie, Ruichen Xia, Haleem Kim, Hyungbin Lim, Jingxu Xie, Takashi Taniguchi, Kenji Watanabe, Michael F. Crommie, Allan H. MacDonald, Feng Wang[‡]. “Two-component exciton condensates in an electron-hole bilayer.” [Nature](#) (2026).
2. **Ruishi Qi**^{†‡}, Qize Li[†], Zuocheng Zhang, Jingxu Xie, Zhiyuan Cui, Takashi Taniguchi, Kenji Watanabe, Michael F. Crommie, Feng Wang[‡]. “Gate-tunable biexcitons in electron-hole-electron trilayers.” Under final editorial processing at [Science](#) (2026).
3. **Ruishi Qi**[†], Qize Li[†], Haleem Kim, Jiahui Nie, Zuocheng Zhang, Ruichen Xia, Zhiyuan Cui, Jiangnan Xiao, Takashi Taniguchi, Kenji Watanabe, Michael F. Crommie, Feng Wang[‡]. “An exciton crystal in a moiré excitonic insulator.” [Nature Physics](#) 22, 514-520 (2026).
4. **Ruishi Qi**[†], Qize Li[†], Zuocheng Zhang, Jiahui Nie, Bo Zou, Zhiyuan Cui, Haleem Kim, Collin Sanborn, Sudi Chen, Jingxu Xie, Takashi Taniguchi, Kenji Watanabe, Michael F. Crommie, Allan H. MacDonald, Feng Wang[‡]. “Competition between excitonic insulators and quantum Hall states in correlated electron-hole bilayers.” [Nature Materials](#) 25, 35-41 (2026).
5. **Ruishi Qi**, Qize Li, Zuocheng Zhang, Sudi Chen, Jingxu Xie, Yunbo Ou, Zhiyuan Cui, David D. Dai, Andrew Y. Joe, Takashi Taniguchi, Kenji Watanabe, Sefaattin Tongay, Alex Zettl, Liang Fu, Feng Wang[‡]. “Electrically controlled interlayer trion fluid in electron-hole bilayers.” [Science](#) 390 (6770), 299-303 (2025).

6. **Ruishi Qi**[†], Andrew Y. Joe^{†‡}, Zuocheng Zhang, Jingxu Xie, Qixin Feng, Zheyu Lu, Ziyu Wang, Takashi Taniguchi, Kenji Watanabe, Sefaattin Tongay, Feng Wang[‡]. "Perfect Coulomb drag and exciton transport in an excitonic insulator." *Science* 388 (6744), 278-283 (2025).
7. **Ruishi Qi**[†], Andrew Y. Joe^{†‡}, Zuocheng Zhang, Yongxin Zeng, Tiancheng Zheng, Qixin Feng, Jingxu Xie, Emma Regan, Zheyu Lu, Takashi Taniguchi, Kenji Watanabe, Sefaattin Tongay, Michael F. Crommie, Allan H. MacDonald, Feng Wang[‡]. "Thermodynamic behavior of correlated electron-hole fluids in van der Waals heterostructures." *Nature Communications* 14, 8264 (2023).
8. **Ruishi Qi**[†], Ruochen Shi[†], Yuehui Li, Yuanwei Sun, Mei Wu, Ning Li, Jinlong Du, Kaihui Liu, Chunlin Chen, Ji Chen, Feng Wang, Dapeng Yu, En-Ge Wang, Peng Gao[‡]. "Measuring phonon dispersion at an interface." *Nature* 599, 399-403 (2021).
9. **Ruishi Qi**[†], Ning Li[†], Jinlong Du, Ruochen Shi, Yang Huang, Xiaoxia Yang, Lei Liu, Zhi Xu, Qing Dai, Dapeng Yu, Peng Gao[‡]. "Four-dimensional vibrational spectroscopy for nanoscale mapping of phonon dispersion in BN nanotubes." *Nature Communications* 12, 1179 (2021).

Additional publications

10. Zuocheng Zhang^{†‡}, **Ruishi Qi**[†], Jingxu Xie, Qize Li, Takashi Taniguchi, Kenji Watanabe, Michael F. Crommie, Feng Wang[‡]. "Orbital dependent Coulomb drag in electron-hole bilayer graphene heterostructures." *Physical Review Letters* 136, 126303 (2026).
11. Zheyu Lu^{†‡}, **Ruishi Qi**[†], Rwik Dutta[†], Jingxu Xie, Jiahui Nie, Ziyu Wang, Yibo Feng, Woonchang Kim, Kenji Watanabe, Takashi Taniguchi, Archana Raja, Mit H. Naik, Steven G. Louie, Feng Wang[‡]. "Nature of emergent moiré excitations in MoSe₂/WS₂ moiré superlattices." *Nano Letters* 26 (12), 4096-4102 (2026).
12. Josue Rodriguez, **Ruishi Qi**, Catherine Xu, Feng Wang, James G. Analytis, Hossein Taghinejad. "Electrically controlled Heat Assisted Magnetic Recording in Intercalated 2D Magnets." arXiv:2605.06645 (2026).
13. Zheyu Lu, Jiahui Nie, Tianle Wang, Rwik Dutta, **Ruishi Qi**, Jingxu Xie, Can Uzundal, Jianghan Xiao, Ziyu Wang, Yibo Feng, Kenji Watanabe, Takashi Taniguchi, James R. Chelikowsky, Archana Raja, Steven G. Louie, Mit H. Naik, Michael P. Zaletel, Feng Wang. "Tunable Interlayer Charge-transfer States in MoSe₂/WS₂ Moiré Superlattices." arXiv:2605.05571 (2026).
14. Bo Han, Ruochen Shi, Huining Peng, Yingjie Lv, **Ruishi Qi**, Yuehui Li, Jingmin Zhang, Jinlong Du, Pu Yu, Peng Gao[‡]. "Electron Microscopy Measurement of Picometer-Level Distortion Induced Local Phonons at a Defect." *Nano Letters* 26 (4), 1449-1454 (2026).
15. Su-Di Chen[†], **Ruishi Qi**, Ha-Leem Kim, Qixin Feng, Ruichen Xia, Dishan Abeysinghe, Jingxu Xie, Takashi Taniguchi, Kenji Watanabe, Dung-Hai Lee, Feng Wang[‡]. "Terahertz electrodynamics in a zero-field Wigner crystal." arXiv:2509.10624 (2025).
16. Trevor Chistolini[†], Ha-Leem Kim, Qiyu Wang, Su-Di Chen, Luke Pritchard Cairns, Ryan Patrick Day, Collin Sanborn, Hyunseong Kim, Zahra Pedramrazi, **Ruishi Qi**, Takashi Taniguchi, Kenji Watanabe, James G. Analytis, David I. Santiago, Irfan Siddiqi, Feng Wang[‡]. "Contactless cavity sensing of superfluid stiffness in atomically thin 4Hb-TaS₂." arXiv:2510.25124 (2025).
17. Su-Di Chen, Qixin Feng, Wenyu Zhao, **Ruishi Qi**, Zuocheng Zhang, Dishan Abeysinghe,

- Can Uzundal, Jingxu Xie, Takashi Taniguchi, Kenji Watanabe, Feng Wang[‡]. "Direct Measurement of Terahertz Conductivity in a Gated Monolayer Semiconductor." [*Nano Letters*](#) 25 (19) 7998-8002 (2025).
18. Chao He, Ze Hua, Peiyi He, **Ruishi Qi**, Yuehui Li, Ruiwen Shao, Weikang Dong, Ruochen Shi, Yeliang Wang, Peng Gao[‡]. "Measurement of far-infrared surface phonon polaritons in AlN nanowires via electron microscope." [*Chinese Physics B*](#) 35, 037901 (2025).
 19. Qixin Feng[†], Can B. Uzundal[†], Ruihan Guo, Collin Sanborn, **Ruishi Qi**, Jingxu Xie, Jianing Zhang, Junqiao Wu, Feng Wang[‡]. "Femtojoule All-optical Nonlinearity for Deep Learning with Incoherent Illumination." [*Science Advances*](#) 11, eads4224 (2025).
 20. Peiyi He[†], Jiade Li[†], Cong Li, Ning Li, Bo Han, Ruochen Shi, **Ruishi Qi**, Jinlong Du, Pu Yu, Peng Gao[‡]. "Strongly confined Mid-infrared to Terahertz Phonon Polaritons in Ultra-thin SrTiO₃." [*Science Advances*](#) 11, eady7316 (2025).
 21. Yue-Hui Li[†], Bo Han[†], Xiao-Long Yang, Rui-Lin Mao, Fa-Chen Liu, Ruo-Chen Shi, **Rui-Shi Qi**, Xiao-Rui Hao, Ning Li, Bing-Yao Liu, Xiao-Mei Li, Jin-Long Du, Ji Chen, Wu Li, Peng Gao[‡]. "Single-dislocation phonons: atomic-scale measurement and their thermal properties." [*Chinese Physics Letters*](#) 42, 066302 (2025).
 22. Yoseob Yoon[‡], Zheyu Lu, Can Uzundal, **Ruishi Qi**, Wenyu Zhao, Sudi Chen, Qixin Feng, Woochang Kim, Mit H. Naik, Kenji Watanabe, Takashi Taniguchi, Steven G. Louie, Michael F. Crommie, Feng Wang[‡]. "Terahertz phonon engineering with van der Waals heterostructures." [*Nature*](#) 631, 771-776 (2024).
 23. Jingxu Xie, Zuocheng Zhang, Haodong Zhang, Vikram Nagarajan, Wenyu Zhao, Ha-Leem Kim, Collin Sanborn, **Ruishi Qi**, Sudi Chen, Salman Kahn, Kenji Watanabe, Takashi Taniguchi, Alex Zettl, Michael F. Crommie, James Analytis, Feng Wang[‡]. "Low Resistance Contact to P-type Monolayer WSe₂." [*Nano Letters*](#) 24 (20), 5937-5943 (2024).
 24. Zuocheng Zhang[†], Jingxu Xie[†], Wenyu Zhao, **Ruishi Qi**, Collin Sanborn, Shaoxin Wang, Salman Kahn, Kenji Watanabe, Takashi Taniguchi, Alex Zettl, Michael Crommie, Feng Wang[‡]. "Engineering correlated insulators in bilayer graphene with a remote Coulomb superlattice." [*Nature Materials*](#) 23, 189-195 (2024).
 25. Ruochen Shi[†], Qize Li[†], Xiaofeng Xu, Bo Han, Ruixue Zhu, Fachen Liu, **Ruishi Qi**, Xiaowen Zhang, Jinlong Du, Ji Chen, Dapeng Yu, Xuetao Zhu, Jiandong Guo, Peng Gao[‡]. "Atomic-scale observation of localized phonons at FeSe/SrTiO₃ interface." [*Nature Communications*](#) 15, 3418 (2024).
 26. Zhenyu Zhang, Tao Wang, Hailing Jiang, **Ruishi Qi**, Yuehui Li, Jinlin Wang, Shanshan Sheng, Ning Li, Ruochen Shi, Jiaqi Wei, Fang Liu, Shengnan Zhang, Xiaoqing Huo, Jinlong Du, Jingmin Zhang, Jun Xu, Xin Rong, Peng Gao, Bo Shen, Xinqiang Wang[‡]. "Probing Hyperbolic Shear Polaritons in β -Ga₂O₃ Nanostructures Using STEM-EELS." [*Advanced Materials*](#) 36, 2204884 (2024).
 27. Jingyuan Yan, Ruochen Shi, Jiake Wei, Yuehui Li, **Ruishi Qi**, Mei Wu, Xiaomei Li, Bin Feng, Peng Gao, Naoya Shibata, Yuichi Ikuhara[‡]. "Nanoscale Localized Phonons at Al₂O₃ Grain Boundaries." [*Nano Letters*](#) 24, 11, 3323-3330 (2024).
 28. Mei Wu, Ruochen Shi, **Ruishi Qi**, Yuehui Li, Jinlong Du, Peng Gao[‡]. "Four-dimensional electron energy-loss spectroscopy." [*Ultramicroscopy*](#) 253, 113818 (2023).
 29. Mei Wu[†], Ruochen Shi[†], **Ruishi Qi**[†], Yuehui Li, Tao Feng, Bingyao Liu, Jingyuan Yan, Xiaomei Li, Zhetong Liu, Tao Wang, Tongbo Wei, Zhiqiang Liu, Jinlong Du, Ji Chen,

- Peng Gao[‡]. “Effects of localized interface phonons on heat conductivity in ingredient heterogeneous solids.” [*Chinese Physics Letters*](#), 40 (3), 036801 (2023).
30. Xiangdong Guo[†], Ning Li[†], Xiaoxia Yang[‡], **Ruishi Qi**, Chenchen Wu, Ruochen Shi, Yuehui Li, Yang Huang, F. Javier García de Abajo, En-Ge Wang, Peng Gao, Qing Dai[‡]. “Hyperbolic whispering-gallery phonon polaritons in boron-nitride nanotubes.” [*Nature Nanotechnology*](#) 18, 529-534 (2023).
 31. Ning Li[†], Ruochen Shi[†], Yifei Li[†], **Ruishi Qi**, Fachen Liu, Xiaowen Zhang, Zhetong Liu, Yuehui Li, Xiangdong Guo, Kaihui Liu, Ying Jiang, Xin-Zheng Li, Ji Chen, Lei Liu, En-Ge Wang, Peng Gao[‡]. “Phonon transition across an isotopic interface.” [*Nature Communications*](#) 14, 2382 (2023).
 32. Yue-Hui Li, **Rui-Shi Qi**, Ruo-Chen Shi, Jian-Nan Hu, Zhe-Tong Liu, Yuan-Wei Sun, Ming-Qiang Li, Ning Li, Can-Li Song, Lai Wang, Zhi-Biao Hao, Yi Luo, Qi-Kun Xue, Xu-Cun Ma, Peng Gao[‡]. “Atomic-scale probing of heterointerface phonon bridges in nitride semiconductor.” [*Proceedings of the National Academy of Sciences*](#) 119 (8), e2117027119 (2022).
 33. Ruixue Zhu[†], Zhexin Jiang, Xinxin Zhang, Xiangli Zhong, Congbing Tan, Mingwei Liu, Yuanwei Sun, Xiaomei Li, **Ruishi Qi**, Ke Qu, Zhetong Liu, Mei Wu, Mingqiang Li, Boyuan Huang, Zhi Xu, Jinbin Wang, Kaihui Liu, Peng Gao[‡], Jie Wang, Jiangyu Li, Xuedong Bai. “Dynamics of Polar Skyrmion Bubbles under Electric Fields.” [*Physical Review Letters*](#) 129 (10), 107601 (2022).
 34. Xiangdong Guo[†], Ning Li[†], Chenchen Wu, Xiaokang Dai, **Ruishi Qi**, Tianyu Qiao, Tuoyi Su, Dandan Lei, Nishuang Liu, Jinlong Du, Enge Wang, Xiaoxia Yang, Peng Gao, Qing Dai[‡]. “Studying Plasmon Dispersion of MXene for Enhanced Electromagnetic Absorption.” [*Advanced Materials*](#) 34 (33), 2270237 (2022).
 35. Yoseob Yoon, Zuocheng Zhang, **Ruishi Qi**, Andrew Y. Joe, Renee Sailus, Kenji Watanabe, Takashi Taniguchi, Sefaattin Tongay, Feng Wang[‡]. “Charge Transfer Dynamics in MoSe₂/hBN/WSe₂ Heterostructures.” [*Nano Letters*](#) 22 (24), 10140-10146 (2022).
 36. Ning Li[†], Xiangdong Guo[†], Xiaoxia Yang[‡], **Ruishi Qi**, Tianyu Qiao, Yifei Li, Ruochen Shi, Yuehui Li, Kaihui Liu, Zhi Xu, Lei Liu, F. Javier García de Abajo, Qing Dai, En-Ge Wang, Peng Gao[‡]. “Direct observation of highly confined phonon polaritons in suspended monolayer hexagonal boron nitride.” [*Nature Materials*](#) 20, 43-48 (2021).
 37. Shilong Zhao[†], Erqing Wang[†], Ebru Alime Üzer, Shuaifei Guo, **Ruishi Qi**, Junyang Tan, Kenji Watanabe, Takashi Taniguchi, Tom Nilges, Peng Gao, Yuanbo Zhang, Hui-Ming Cheng, Bilu Liu, Xiaolong Zou, Feng Wang[‡]. “Anisotropic moiré optical transitions in twisted monolayer/bilayer phosphorene heterostructures.” [*Nature Communications*](#) 12, 3947 (2021).
 38. Weikang Dong[†], **Ruishi Qi**[†], Tiansheng Liu, Yi Li, Ning Li, Ze Hua, Zirui Gao, Shuyuan Zhang, Kaihui Liu, Jiandong Guo, Peng Gao[‡]. “Broad-Spectral-Range Sustainability and Controllable Excitation of Hyperbolic Phonon Polaritons in α -MoO₃.” [*Advanced Materials*](#) 32, 2002014 (2020).
 39. Yuehui Li, **Ruishi Qi**, Ruochen Shi, Ning Li, Peng Gao[‡]. “Manipulation of surface phonon polaritons in SiC nanorods.” [*Science Bulletin*](#) 65 (10), 820-826 (2020).
 40. Yue-Hui Li[†], Mei Wu[†], **Rui-Shi Qi**, Ning Li, Yuan-Wei Sun, Cheng-Long Shi, Xue-Tao Zhu, Jian-Dong Guo, Da-Peng Yu, Peng Gao[‡]. “Probing lattice vibrations at SiO₂/Si surface

and interface with nanometer resolution." [*Chinese Physics Letters*](#) 36 (2), 026801 (2019).

41. **Ruishi Qi**[†], Renfei Wang[†], Yuehui Li, Yuanwei Sun, Shulin Chen, Bo Han, Ning Li, Qing Zhang, Xinfeng Liu, Dapeng Yu, Peng Gao[‡]. "Probing far-infrared surface phonon polaritons in semiconductor nanostructures at nanoscale." [*Nano Letters*](#) 19 (8), 5070-5076 (2019).

Honors and Fellowships

- 2026 – 2029 Stanford Science Fellow, Stanford University
2026 Jackson C. Koo Award in Condensed Matter, University of California, Berkeley
2026 Alex Zettl Graduate Student Award in Physics, University of California, Berkeley
2023 – 2024 Kavli ENSI Graduate Student Fellow, Kavli Energy NanoScience Institute
2020 Weiming Scholar, Peking University
2018, 2019 National Scholarship (China)

Research Experience

- 2021 – 2026 **Graduate Student Researcher**, Department of Physics, UC Berkeley
Ultrafast nano-optics group (PI: Prof. Feng Wang)
 - Established electrically tunable van der Waals electron-hole bilayers as a platform for correlated exciton phases, including interlayer excitons, trions, biexcitons, exciton condensates, and exciton crystals.
 - Developed optical spectroscopy approaches for measuring compressibility, Coulomb drag, and transport properties in strongly interacting two-dimensional systems.
 - Designed and performed low-temperature, high-magnetic-field optical and transport experiments down to 10 mK and up to 12 T.

2018 – 2021 **Undergraduate Research Assistant**, Peking University
Electron microscopy laboratory (PI: Prof. Peng Gao)
 - Developed electron energy-loss spectroscopy techniques combining nanometer spatial resolution, high momentum resolution, and milli-electron-volt energy resolution.
 - Investigated local lattice dynamics at heterointerfaces and nanostructures.

Teaching and Mentoring

- 2020 – 2021 **Graduate Student Instructor**, Department of Physics, UC Berkeley
Physics 7A: Physics for Scientists and Engineers (Fall 2020, Spring 2021)
Led discussion and laboratory sections; graded homework and exams; held office hours.
2023 – 2024 **Undergraduate Research Mentor**, Berkeley Physics Innovators Initiative (Pi²)
Mentored Berkeley undergraduate researchers (selective funded research program).
2022, 2025 **Undergraduate Research Mentor**, Berkeley Physics International Education (BPIE)
Mentored visiting undergraduate researchers.

Talks and Presentations

- 2026 "Correlated exciton physics in strongly coupled electron-hole bilayers", APS March Meeting, Denver
2026 "Excitonic insulators and condensates in electron-hole bilayers" (Poster), New Frontiers in Nanoscale Materials 2026 Symposium
2025 "Interlayer excitons and trions in electron-hole bilayers", Berkeley Condensed Matter Seminar

- 2024 "Perfect Coulomb drag and exciton transport in an excitonic insulator", APS March Meeting, Minneapolis
- 2024 "Correlated interlayer excitons and trions in electron-hole bilayers" (Poster), Kavli ENSI 10th Anniversary Symposium